

Does Digital Assistance Deepen or Disrupt Player Experience in Virtual Tabletop Role-Playing Games?



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Abstract

The rapid digitisation of Tabletop Role-Playing Games (TTRPGs) has introduced complex automation into traditional face-to-face play, creating hybrid “phygital” gaming environments. While digital assistance tools reduce the mathematical and administrative burdens associated with gameplay, their integration into physical spaces risks fracturing social presence and narrative immersion through isolating, “head-down” interaction paradigms. This dissertation investigates whether digital assistance tools deepen or disrupt immersion in in-person TTRPG sessions by altering players’ cognitive load.

To explore this, a bespoke, dual-display software artefact was developed using the Unity Engine. Designed through the lens of Cognitive Load Theory, the tool features a private Game Master dashboard for automating spatial measurements and token management, alongside a shared player-facing screen intended to centralise visual attention. The system was evaluated through a mixed-methods user study, triangulating quantitative workload data from the NASA Task Load Index (NASA-TLX) with qualitative thematic analysis of semi-structured player interviews.

The findings indicate that the digital artefact successfully eradicated extraneous cognitive load, evidenced by exceptionally low Frustration (2.6/10) and high Performance (8.8/10) scores, allowing players to redirect their mental effort toward germane, strategic role-play. Furthermore, the shared display fostered “head-up” social dynamics, significantly deepening systemic and imaginative immersion during combat downtime. However, the study also identified a paradox of the hybrid space: while the software protected narrative flow, the physical hardware invited external digital distractions. Ultimately, this research concludes that digital assistance effectively enhances the tabletop experience when designed to centralise shared attention, offering critical UI/UX design guidelines for the future development of hybrid TTRPG software.

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Chapter 1

Introduction

1.1 Motivation

1.1.1 The Digital Transformation of TTRPGs

Tabletop Role-playing (TTRPGs) are unique collaborative storytelling experiences that merge structured rule systems with unbound creative improvisation. Historically, the medium has been anchored in the physical world where players gather around a table utilising pen, paper, physical dice, and rulebooks to construct a shared imaginative space (Cover, 2014). However, over the last decade, and significantly accelerated by the global shift towards remote interaction, TTRPGs have undergone a profound digital transformation. Platforms such as D&D Beyond, Roll20, and FoundryVTT have introduced high levels of automation, transforming the way players interact with the game’s underlying mechanics.

1.1.2 Cognitive Load in Tabletop Gaming

The primary motivation behind the digitisation is the reduction of cognitive load. Modern TTRPG systems, such as Dungeons & Dragons 5th Edition, require players and Game Masters (GMs) to juggle complex Mathematics, spatial positioning, and vast amounts of character information. Digital tools aim to alleviate this “extraneous cognitive load” (Sweller, 1988), theoretically freeing up mental resources for role-playing and strategic thinking.

1.2 The Hybrid Environment and Social Architecture

Moreover, the social architecture of TTRPGs heavily relies on a conceptual boundary where players suspend disbelief and immerse

themselves in the shared fiction. In traditional face-to-face play, this is maintained through constant eye contact, reading non-verbal cues, and a shared focus on physical artifacts in the centre of the table. When digital assistance is introduced to an in-person setting, creating a ‘phygital’ (physical-digital) or hybrid gaming space, the dynamic shifts. Players must divide their attention between the shared physical tabletop and the private digital space of their screens. This ‘head-down’ interaction paradigm risks fracturing the shared narrative focus. While extensive research has evaluated purely remote play platforms, there is a distinct lack of studies regarding how these digital tools affect the intimate, face-to-face dynamics of in-person sessions.

1.2.1 The Paradox of Automation

However, automation is a double-edged sword. Systems designed to assist human operators can inadvertently distance them from the task, leading to a loss of engagement and situational awareness (Bainbridge, 1982). In the context of a TTRPG, if a digital tool automates the rolling of dice and calculations of modifiers, does it remove the tactile satisfaction and dramatic tension inherent to physical play? Does the presence of a screen in a face-to-face setting disrupt the shared focus required to maintain narrative immersion? This project is motivated by the need to understand this delicate balance: determining where digital assistance transitions from a helpful cognitive aid to a disruptive barrier in hybrid gaming environments.

1.3 Aims and Objectives

1.3.1 Project Aims and Core Objectives

The primary aim of this project is to investigate and critically evaluate how the use of digital assistance tools affects the cognitive workload and immersive engagement during in-person, hybrid Virtual Tabletop Role-Playing Game (VTT RPG) sessions.

To achieve this aim, the following objectives have been established:

1. Literature Review: To critically evaluate existing literature surrounding cognitive load theory, human-automation interaction, and immersion within physical and digital gaming environments.

2. **Artifact Design and Development:** To design and implement a Unity-based digital assistance tool tailored for in-person TTRPG play, featuring a dual-display system separating GM and player information.
3. **Experimental Study Design:** To design and conduct an ethically approved, mixed-methods study where participants utilise the developed artifact during live tabletop sessions.
4. **Data Collection and Analysis:** To measure cognitive workload using the standardised NASA-TLX assessment and capture experiential data via semi-structured interviews, subsequently analysing this data through statistical and thematic analysis.
5. **Evaluation and Recommendations:** To synthesise the findings, drawing conclusions on the impact of digital tools on TTRPG immersion, and to propose design guidelines for future hybrid tabletop software.

1.3.2 Methodological Rationale

By adopting a mixed-methods approach, this project bridges principles from Human-Computer Interaction (HCI) and Game Studies. While quantitative metrics offer an objective measure of cognitive exertion, the qualitative interviews ensure that the nuanced, subjective nature of role-playing immersion, which cannot be easily quantified, is adequately captured. Ultimately, these insights will contribute to the ongoing academic discourse surrounding technology-mediated social play.

1.4 Dissertation Structure

1.4.1 Chapter Overview

This dissertation is structured to document the complete lifecycle of the project, from theoretical grounding to a final evaluation.

- Chapter 2 (Literature Review) explores the theoretical background of the study, examining cognitive load, theories of immersion, and existing research on VTTs.
- Chapter 3 (Requirements Analysis) outlines the functional and non-functional requirements for the software artifact, alongside the methodology for evaluation.

- Chapter 4 (Design and Methodology) details the design of the Unity artifact, the UI/UX decisions, and the methodological framework for the mixed-methods user study.
- Chapter 5 (Implementation) provides a technical breakdown of the software development process, highlighting key programming challenges and solutions in the Unity Engine.
- Chapter 6 (Results and Discussion) presents the quantitative NASA-TLX data alongside the qualitative thematic analysis of the player interviews, discussing these findings in the context of the original research question.
- Chapter 7 (Conclusion) summarises the project’s achievements, acknowledges its limitation, and suggests avenues for future work in the domain of tabletop gaming.

1.4.2 Research Outcome

Ultimately, this dissertation seeks to provide a comprehensive understanding of how digital tools can be harmoniously integrated into physical gaming spaces. By addressing both the technical challenges of developing a hybrid assistance tool and the human-computer interaction elements of player experience, the research outcomes aim to offer actionable guidance for developers in this rapidly expanding domain. The following chapter will lay the theoretical foundation for this investigation by reviewing current literature on cognitive load, automation, and immersion.

Chapter 2

Literature Review

This chapter critically examines the existing academic literature surrounding tabletop role-playing games (TTRPGs), cognitive load, and the human-computer interaction (HCI) to establish a theoretical foundation for this project. The review is structured chronologically and thematically, beginning with the historical evolution of TTRPGs and their recent digital transformation. It then explores many psychological frameworks, specifically Cognitive Load Theory and theories of immersion, to understand how players process information and maintain narrative engagement. Finally, the chapter reviews current research on automation, hybrid gaming spaces, and the methodological tools used to measure subjective player experiences, ultimately identifying the critical research gap this dissertation addresses.

2.1 The Evolution of TTRPGs and the Digital Shift

2.1.1 The Traditional Physical Experience

Tabletop role-playing games are fundamentally collaborative narrative experiences that blend cooperative storytelling, rigid rule systems, and social interaction (Cover, 2014). Traditionally played face-to-face, TTRPGs have historically relied entirely on physical artifacts: paper character sheets that track statistics, physical rulebooks for mechanical arbitration, and polyhedral dice to introduce elements of chance and resolve uncertainty. In this traditional setup, the cognitive and administrative burden falls entirely on the players and the Game Master (GM). These physical components are not merely administrative tools; they serve as tactile anchors that help players ground themselves within the shared imaginative space, fostering a sense of community and collaborative problem-solving (Bowman, 2010).

2.1.2 The Rise of Virtual Tabletops

Over the past decade, TTRPGs have undergone a significant digital transformation. This shift was initially driven by the desire to connect players across geographical distances but was massively accelerated by the global COVID-19 pandemic, which necessitated remote play. As a result, Virtual Tabletop (VTT) platforms such as Roll20, FoundryVTT, Owlbear Rodeo, and companion applications like D&D Beyond emerged as dominant forces in the hobby. These platforms digitise the traditional digital artifacts automating mechanical elements such as dice rolls, modifier calculations, spell area-of-affect templates, and character statistic management.

2.1.3 The Cost of Streamlining

While these digital assistance tools promise enhanced accessibility and a reduced administrative burden, their impact on the core experiential qualities of the game remains a subject of debate. Traditional physical play requires players to actively engage with the rules. Adding modifiers to a physical dice roll requires a working knowledge of the character's capabilities. In contrast, digital tools often reduce this interaction to a single button click. While this streamlines combat and reduces mathematical errors, it raises questions about whether this automation distances the player from their character's mechanics.

2.1.4 The Emergence of Hybrid Play

Furthermore, the majority of VTT development and associated research has focused almost exclusively on fully remote play. However, as in-person play has resumed, players are increasingly bringing these digital tools back to the physical table, creating hybrid gaming environments. The impact of this digital assistance on core tabletop qualities, such as imaginative immersion, creative engagement, and narrative flow in a face-to-face setting, remains largely unexplored. Understanding this shift requires looking beyond the software's functionality and examining how digital interfaces alter the psychological state of the players.

2.2 Theories of Immersion, Presence, and Flow

2.2.1 The Dimensions of Immersion, Presence, and Flow

To understand whether digital assistance disrupts or deepens the tabletop experience, it is first necessary to define “immersion,” a term often used colloquially but possessing specific academic dimensions. Immersion in gaming is generally understood as the degree of involvement a player has with the virtual or imaginative environment.

A foundational framework for understanding this is the SCI model, which divides immersion into three components: Sensory, Challenge-based, and Imaginative (Ermi and Mäyrä, 2005). Traditional TTRPGs rely almost entirely on Imaginative immersion where players use spoken descriptions and their “theatre of the mind” to visualise the game world and empathise with their characters. Digital tools and VTTs, however, introduce a heavier reliance on Sensory immersion through digital maps, automated soundscapes, and visual tokens. A critical question arises as to whether this increase in sensory stimuli supports the imaginative experience or overrides it, making the game feel more like a traditional video game and less like a collaborative storytelling experience.

2.2.2 Perserving the Flow State

Closely related to immersion is the theory of Flow, defined as a state of deep absorption in an activity where the user’s skill level perfectly matches the challenge presented (Csikszentmihalyi, 2000). In TTRPGs, flow is achieved when narrative pacing is smooth and mechanical resolution (like rolling dice) feels intuitive and seamless. When players have to stop and manually calculate complex maths, flow can be broken. Digital assistance theoretically protects this flow state by automating the math. However, if a digital tool is poorly designed, clunky to navigate, or cause software errors, it introduces a new type of friction that shatters flow entirely.

2.2.3 Social Presence and the ‘Magic Circle’

Furthermore, “Presence”, the psychological feeling of “being there” in the game world, is heavily mediated by social interactions in TTRPGs. The shared physical space of a tabletop anchors the players. Introducing screens can create a barrier to this social presence, splitting a player’s attention between the physical table and the digital interface.

2.3 Cognitive Load Theory in Gaming

2.3.1 Managing Working Memory

Understanding how players process game mechanics requires an examination of the Cognitive Load Theory (CLT). CLT proposes that human working memory is limited and can easily become overwhelmed if a task demands too much mental effort (Sweller, 1988). CLT divides cognitive load into three categories: Intrinsic load (the inherent complexity of the task), Extraneous load (unnecessary complexity introduced by poor design or presentation), and Germane load (the productive cognitive effort used to learn, strategise, and engage with the task).

Optimal performance and engagement occur when extraneous load is minimised, allowing mental resources to be dedicated to the germane load (Paas Et al., 2003). In TTRPGs, calculating multiple overlapping buffs, debuffs, and dice modifiers represents a high extraneous load. Digital tools are designed specifically to eliminate this burden.

2.3.2 Asymmetrical Cognitive Demands: GM vs. Player

However, an important distinction must be made between the Game Master (GM) and the players. GMs bear an immense intrinsic cognitive load: they must manage multiple non-player characters (NPCs), track narrative continuity, arbitrate rules, and react to unpredictable player choices. For GMs, digital automation is almost universally beneficial, as it prevents cognitive overload. Players, on the other hand, manage only a single character. If a digital tool automates all their mechanics (such as automatically calculating hit points and attack bonuses), it may inadvertently eliminate their

germane load. By removing the need to actively interact with their character’s statistics, players might become passive consumers of the game rather than active participants, thereby reducing their overall engagement.

2.3.3 Critiquing the Transfer of CLT to Role-Playing

While Cognitive Load Theory provides a robust vocabulary for discussing mental effort, its direct transfer from instructional design to collaborative storytelling must be critically examined. Originally formulated to optimise classroom learning, CLT assumes that the ultimate goal of an interface is absolute cognitive efficiency (Mayer and Fiorella, 2022). However, game studies scholars have argued that gaming inherently relies on “productive friction”—the deliberate introduction of challenges that make the experience engaging rather than effortless (Deterding, 2015).

In industrial or educational contexts, calculating mathematics is a barrier to the primary task. In a TTRPG, however, the act of rolling physical dice and adding modifiers is often cited by players as a tactile ritual that builds dramatic tension (Bowman, 2018). Therefore, directly applying CLT to TTRPGs risks an over-simplification: assuming that because a digital tool reduces extraneous mathematical load, it automatically improves the player’s experience. This assumption fails to account for whether the physical administration of the game is actually “extraneous,” or if it is a deeply ingrained, ritualistic component of tabletop immersion. A critical examination is required to determine whether automating these mechanics strips away the tangible engagement that originally defined the medium.

2.4 Automation and Human-Computer Interaction

2.4.1 The Ironies of Automation in Play

The paradox of automating tabletop mechanics can be further understood through “Ironies of Automation” (Bainbridge, 1982). Bainbridge established that systems intended to reduce human operator workload often increase cognitive demands in unexpected

ways or degrade the operator’s fundamental skills. In TTRPGs, players who rely entirely on an app to calculate their abilities may fail to learn the game’s underlying rules. Consequently, when the automation fails, such as when a complex, ambiguous narrative situation arises that the software cannot account for, the player lacks the foundational knowledge required to intervene manually.

2.4.2 Patterns of Human-Automation Interaction

Further exploration of human-automation interaction identifies distinct patterns of user behavior: specifically misuse, disuse, and abuse (Parasuraman and Riley, 1997). Applied to the VTT context, “misuse” occurs when players blindly trust the digital tool’s output, failing to notice when a modifier was applied incorrectly due to a software limitation. Conversely, “disuse” happens when players abandon a potentially helpful digital tool because an early bug or clunky interface breaks their trust or disrupts the narrative flow.

2.4.3 Balancing Automation and Agency

Therefore, automation in TTRPGs is not a neutral addition; it fundamentally alters the player’s relationship with the game’s mechanics. The ideal digital assistance tool must strike a balance: it should automate tedious administrative tasks while requiring enough manual input from the player to maintain their situational awareness and active engagement with the game state.

2.4.4 Contextualising Industrial Automation for Collaborative Play

The frameworks established were developed within high-stakes industrial environments, such as aviation and nuclear operation, where automation failure results in catastrophic physical consequences (Bainbridge, 1982; Parasuraman and Riley, 1997). Applying these models to the leisure context of a TTRPG requires contextual translation.

In a TTRPG, an “automation error”, such as a digital map glitching or a VTT miscalculating a modifier, does not result in physical danger, but rather a rupture of the ‘magic circle’ (Huizinga, 1950). The stakes are entirely narrative and social. When digital tools

in a hybrid setting fail or cause confusion, they force the players to break character and address the software as users rather than as role-players. Thus, while the patterns of “misuse” and “disuse” remain highly relevant, the consequence of poor automation in TTRPGs is not operational failure, but rather the rapid deflation of imaginative immersion. Existing literature frequently overlooks this distinction, treating gaming software usability as a purely functional metric rather than an emotional and narrative vulnerability.

2.5 Hybrid Play Spaces

2.5.1 The ‘Head-Down’ Interaction Paradigm

While much recent research has focused on purely remote digital gaming (facilitated via internet calls and VTTs), the return to in-person play has created a hybrid, or “phygital,” environment. Players sit physically at the same table but interact via individual laptops, tablets, or smartphones.

Investigations into board game play have demonstrated that physical artifacts and face-to-face interactions create unique social affordances that digital games struggle to replicate (Rogerson Et al., 2018). Physical components naturally direct attention to the center of the table, creating a shared focal point. Conversely, introducing personal screens introduces a “head-down” dynamic. Instead of looking at the GM or other players to read social cues and engage in role-play, players may focus downward on their individual interfaces.

2.5.2 The Limitations of AR and VR Tabletop Research

Recent attempts to bridge the physical-digital divide in tabletop gaming have frequently leaned toward Augmented Reality (AR) and Virtual Reality (VR) solutions. Several studies have explored using AR headsets to project 3D digital game boards onto physical tables, aiming to merge digital automation with physical presence. However, while these studies are technologically pioneering, their findings cannot be directly applied to standard hybrid in-person play.

The fundamental flaw in transferring AR/VR research to the standard ‘phygital’ tabletop is the isolation of the visual field. Head-mounted displays inherently enclose the user. Even in see-through AR, players are viewing their own distinct projection rather than a unified, shared physical artifact. This isolates the players perceptually, damaging the shared gaze and micro-expressions critical to collaborative role-play (Gugenheimer Et al., 2018). A shared secondary monitor, conversely, acts much like a traditional physical game board: it is a singular, objective focal point that exists identically for all players in the room. Therefore, to understand the true impact of digital assistance in modern hybrid TTRPGs, research must pivot away from isolating headset technologies and focus critically on shared, screen-based environmental augmentations.

2.5.3 UI/UX Challenges in Shared Spaces

While digital tools have modernised TTRPGs, questions have been raised regarding how these tools reshape the intimacy of the physical table (Švelch, 2022). Furthermore, research indicates that while digital overlays can enhance a player’s understanding of complex game environments, they can also introduce cognitive distractions that fragment attention (Žaba Et al., 2025). Designing for a hybrid space therefore requires a deep consideration of UI/UX design. Interfaces must be clean and easily readable at a glance so that players can quickly retrieve the information they need and return their attention to the shared physical and social space.

2.6 Methodological Approaches to Measuring Experience

2.6.1 Quantitative Measurement of Workload

Measuring a subjective experience like “immersion” alongside a quantitative metric like “cognitive load” requires robust, validated methodologies. The NASA Task Load Index (NASA-TLX) remains the gold standard for assessing perceived workload (Hart, 2006). It measures workload across six dimensions: mental demand, physical demand, temporal demand, performance, effort, and frustration.

While originally designed for aviation and industrial contexts, NASA-TLX has been successfully adapted to gaming environ-

ments. However, workload metrics alone cannot capture the full spectrum of user experience; a high mental demand might indicate frustration (extraneous load) or deep, enjoyable strategic engagement (germane load) (Wiebe Et al., 2014). Therefore, NASA-TLX provides an excellent quantitative baseline but must be contextualised.

2.6.2 The Necessity of Qualitative Context

To provide this context, qualitative methods are essential. The utilisation of mixed methods to examine player involvement has shown that post-session interviews reveal nuanced experiential dimensions, such as emotional investment and social friction, that standardised questionnaires often miss (Iacovides Et al., 2015). Similarly, reflective narratives have proven highly valuable in exploring subjective digital game experiences (Poels Et al., 2007). By combining the quantitative rigor of NASA-TLX with the rich, subjective data gathered from semi-structured interviews, a comprehensive picture of how digital tools impact the TTRPG experience can be formed.

2.7 Summary and Research Gap

2.7.1 Synthesising the Literature

The literature demonstrates that while TTRPGs have embraced digital transformation, the theoretical implications of this shift are complex. Cognitive Load Theory suggests that digital tools should theoretically benefit players by reducing extraneous mathematical burdens. However, theories of automation and immersion warn that over-automation can alienate players from the game mechanics, disrupt flow, and damage the imaginative and social presence required for a successful session.

2.7.2 Addressing the Hybrid Research Gap

Despite growing interest in digital TTRPGs, the current academic discourse exhibits a distinct research gap. Most existing studies focus either on fully automated video games or purely remote VTT play. There is a lack of empirical research examining the hybrid space, where in-person players utilise digital assistance tools at a physical table. Furthermore, few studies integrate both cogni-

tive workload measurement and qualitative immersion assessment within a unified framework.

This project addresses this gap. By developing a bespoke Unity-based digital assistance tool and evaluating its use in live, face-to-face sessions through a mixed-methods approach, this research will provide evidence-based insights into whether digital tools deepen the core experiential qualities of tabletop role-playing, or disrupt them.

Chapter 3

Requirements Analysis

The literature review established that digital assistance in Tabletop Role-Playing Games (TTRPGs) must carefully balance the reduction of extraneous cognitive load with the preservation of imaginative immersion and social presence. To investigate this balance empirically, an artifact must be developed and evaluated. During the initial planning phases, it was determined that attempting to replicate complex character management systems would be redundant given the widespread adoption of robust platforms like D&D Beyond. Therefore, this project scopes its technical development to focus strictly on the shared physical space of the tabletop. This chapter outlines the target use case, details the functional and non-functional requirements for a Unity-based dual-display assistance and logging tool, specifies the environment, and establishes the evaluation methodology.

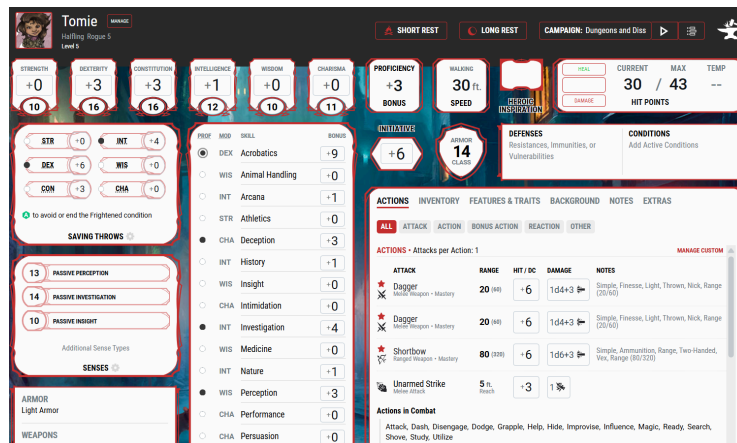


Fig. 1. A brief snippet of a D&D Beyond Character Sheet

3.1 Target Audience and Use Case

3.1.1 User Personas and Cognitive Profiles

User Personas and Cognitive Profiles To accurately gather requirements, the needs of the two distinct tabletop roles must be addressed:

- The Overwhelmed Game Master: The GM bears the highest intrinsic cognitive load. They must track narrative flow, enemy statistics, environmental hazards, and player conditions simultaneously. Their primary requirement is frictionless automation, tools that allow them to retrieve rules and measure distances instantly without pausing the game.
- The Distracted Player: The player bears a lower intrinsic load but is highly susceptible to environmental distraction. During combat “downtime” (when it is not their turn), players frequently disengage. Their primary requirement is a shared visual anchor that keeps them situated within the game state and prevents them from tabbing out to non-game-related applications.

3.1.2 Primary Demographic

The primary target audience consists of players and Game Masters (GMs) engaging in face-to-face, in-person gameplay. While players may manage their individual characters using personal devices (e.g., via D&D Beyond on a smartphone), the shared table often lacks a centralised digital focal point, leading to a fragmented “head-down” experience.

3.1.3 The Shared Digital Hub

The intended use case for the developed artefact involves a single computing device (a laptop) utilised by the GM, connected to a secondary display (a monitor or television) facing the players. The Unity application acts as a central hub for the table. The GM uses their private screen to quickly access rules and manage the flow of information, while pushing relevant rule clarifications or visual aids to the secondary player-facing screen to keep the group’s attention focused up and toward the center of the table.

3.2 Functional Requirements

3.2.1 Defining System Capabilities

Functional requirements define the specific behaviors and capabilities the system must possess to facilitate gameplay and gather research data. The following requirements were established to support the hybrid gaming space:

Table 1. Functional Requirements of the Digital Assistance Artefact

ID	Feature	Requirement	Justification
FR1	Rule Reference Database	The system must contain a searchable, easily accessible database of common game mechanics, status effects, and rules.	Prevents the disruption of narrative flow caused by manually searching through physical rulebooks or multiple browser tabs.
FR2	Dual-Display Architecture	The application must support a multi-monitor setup: a private dashboard on the primary screen, and a restricted public interface on the secondary screen.	Facilitates the hybrid “phygital” space by keeping Game Master secrets hidden while providing a shared screen for the players.
FR3	Information Casting	The GM must be able to select specific rules, images, or status effects from their dashboard and “cast” them to the player display.	Creates a shared visual anchor for the table, pulling players’ attention up and keeping them engaged during combat downtime.
FR4	Spatial Measurement Tool	The software must include digital tools for calculating distance (rulers) and spell dimensions (Area-of-Effect spheres).	Directly automates the extraneous cognitive load of calculating complex tabletop grid mathematics, speeding up combat resolution.

3.3 Non-Functional Requirements

3.3.1 Defining Quality Standards

Non-functional requirements specify the quality attributes and usability standards the system must meet to ensure it does not negatively impact the user experience.

Table 2. Non-Functional Requirements of the Digital Assistance Artefact

ID	Feature	Requirement	Justification
NFR1	Usability & Clarity	The UI must be minimalist, logically organised, and highly legible at a distance for the player-facing screen.	Complex interfaces increase extraneous cognitive load. A clean UI ensures quick information retrieval without frustrating the user.
NFR2	Performance & Responsiveness	The application must respond to inputs, rule searches, and screen-casting commands with minimal latency.	Delayed feedback would artificially inflate the “Temporal” and “Frustration” dimensions of the users’ NASA-TLX scores during testing.
NFR3	Thematic Visual Cohesion	The UI design should employ a visual aesthetic that subtly aligns with the fantasy genre (e.g., dark mode, appropriate typography).	Avoids sterile, overly-corporate design languages that might break the ‘magic circle’ of play and damage imaginative immersion.

3.4 Software and Hardware Environment

3.4.1 Development Platform

To achieve these requirements, the artifact will be developed using the Unity Engine alongside C# scripting. During the scoping

phase, web-based frameworks (such as React or Vue) were considered. However, web browsers handle secondary-window popouts inconsistently, and syncing data between two separate browser windows in real-time often requires local server hosting or web-sockets, introducing potential latency.

Unity was deliberately selected because of its native rendering pipeline, which supports multi-camera, multi-display output from a single, locally executed instance. This entirely eliminates network latency (satisfying NFR2) and guarantees that the GM dashboard and Player screen remain perfectly synchronised. Furthermore, Unity’s robust UI Toolkit and 3D physics engine provide the ideal foundation for developing the spatial measurement tools.

3.4.2 Target Deployment Hardware

The software will be built as a standalone executable for the Windows operating system. The hardware environment required for deployment and testing consists of a standard modern laptop equipped with a secondary video output (e.g., HDMI or Display-Port) and an external monitor or television. This setup reflects the most accessible and common configuration utilised by players integrating digital tools into physical tabletop spaces.

3.5 Evaluation and Testing Strategy

3.5.1 Technical Verification

The evaluation of the artefact encompasses both technical system testing and a human-centered user study, ensuring the software meets both operational standards and the project’s academic objectives.

System Testing: Iterative testing will be conducted during development to verify that the dual-display functionality correctly routes the appropriate UI panels to the designated monitors without graphical artifacting. Furthermore, the logging system (FR4) will be tested to ensure JSON files are formatted correctly and timestamps are accurately recorded without degrading software performance.

3.5.2 Mixed-Methods User Study

User Evaluation: Once the prototype is functionally complete, it will be evaluated through an ethically approved, mixed-methods user study. Participants will play through a standardised TTRPG scenario utilising the software alongside their standard character management tools (e.g., D&D Beyond).

- Quantitative Evaluation: Following the session, participants will complete the NASA Task Load Index (NASA-TLX) questionnaire to quantify their perceived cognitive, physical, and temporal workloads.
- Qualitative Evaluation: Participants will engage in semi-structured interviews designed to capture experiential data regarding their sense of immersion, narrative flow, and their subjective feelings on how the shared display influenced table focus.
- Log Analysis: The interaction logs generated by the software will be cross-referenced with the qualitative data to determine if high tool-interaction frequency correlates with feelings of disconnection or enhanced flow.

Chapter 4

Design and Methodology

This chapter details the design process of the Unity-based digital assistance artefact and the methodological framework established to evaluate its impact on player immersion and cognitive load. The chapter is divided into two primary sections: software design and research methodology. The software design section outlines the system architecture, dual-display logic, and User Interface (UI) decisions aimed at minimising extraneous cognitive load. The research methodology section details the mixed-methods study design, ethical considerations, data collection procedures involving the NASA-TLX, and the strategy for qualitative thematic analysis.

4.1 Software Development Methodology

4.1.1 An Agile Approach

An iterative, Agile-inspired development methodology was adopted for the creation of the software artefact. Given the highly subjective nature of User Experience (UX) in tabletop gaming, a rigid Waterfall approach was deemed unsuitable. Instead, development was broken down into two-week sprint cycles focusing on core functionalities: first, establishing the dual-camera rendering pipeline in Unity; second, developing the underlying rules database and search functionality; and finally, implementing the session logging system. This iterative approach allowed for rapid prototyping and continuous self-evaluation against the cognitive load theories established in the literature review, ensuring the interface remained unobtrusive.

4.2 System Architecture and Dual-Display Design

4.2.1 The Multi-Camera Display Pipeline

The core architectural challenge of the artefact was facilitating a shared physical space without forcing players to crowd around a single small screen. To achieve this, the system was designed within the Unity Engine using a dual-display architecture.

The application utilises Unity’s multi-display rendering capabilities to drive two separate cameras.

- The Primary Display (GM Dashboard): Rendered on the host laptop, this private interface contains the search bar, the complete rules database, and the control mechanisms for casting information.
- The Secondary Display (Player View): Rendered on a connected external monitor or television facing the players, this display is intentionally sparse. It acts as a digital centerpiece for the table, displaying only the information explicitly pushed to it by the GM, such as a specific rule clarification, an image, or a combat condition (e.g., “Blinded” or “Grappled”).

4.2.2 Localised Data and Background Logging

The rules data is structured locally, allowing the application to function entirely offline. This eliminates the latency and immersion-breaking disruptions commonly associated with web-based searches and slow internet connections. Furthermore, a background logging manager script was implemented to listen for specific UI events (such as button clicks or search queries), appending these events alongside a timestamp to a local JSON file.

4.2.3 Hardware Accessibility and Realistic Deployment

A critical design decision regarding the system architecture was ensuring the hardware requirements mirrored realistic, accessible TTRPG environments. While advanced tabletop setups might employ custom-built digital touch-tables embedded in physical furniture, these are prohibitively expensive and rare. Therefore,

the artefact was deliberately constrained to operate on a standard, consumer-grade laptop connected to a generic secondary monitor or television via HDMI. By designing the dual-display architecture around hardware that the average gaming group already owns, the resulting data regarding immersion and usability becomes highly ecologically valid and generalisable to the broader TTRPG community.

4.3 User Interface (UI) and User Experience (UX) Design

4.3.1 Mitigating Extraneous Load

The UI was designed strictly through the lens of Cognitive Load Theory (Sweller, 1988). Every interface element was evaluated based on whether it contributed to extraneous load (unnecessary clutter) or germane load (useful information aiding the game).

The GM dashboard features a dark-mode aesthetic. This serves a dual purpose: it aligns thematically with the fantasy TTRPG genre, preserving the “magic circle” of play, and it reduces eye strain during long tabletop sessions. The search functionality was prioritised at the top of the visual hierarchy, allowing the GM to type keywords and instantly retrieve rules. This rapid retrieval system is designed to preserve Csikszentmihalyi’s state of “Flow” by reducing the time spent pausing the narrative to consult physical rulebooks.

4.3.2 Fostering ‘Head-Up’ Social Dynamics

Conversely, the Player View on the secondary monitor was designed to promote “head-up” interaction. By placing the shared display centrally, players are encouraged to look up and across the table at one another, mitigating the isolating “head-down” effect caused by individual smartphones or laptops. The typography on the player display was scaled for readability at a distance of several meters, ensuring that all participants can read shared information without breaking their physical posture.

4.4 Research Methodology and Study Design

4.4.1 A Mixed-Methods Framework

To evaluate whether this software design successfully deepens or disrupts immersion, a mixed-methods research design was selected. Subjective experiences like immersion and narrative flow cannot be entirely quantified, while cognitive workload requires standardised metrics to be compared reliably.

Therefore, a convergent parallel design was utilised, collecting both quantitative data (via the NASA Task Load Index and objective interaction logs) and qualitative data (via semi-structured interviews) sequentially during a single experimental session. This triangulation of data sources provides a holistic view of the human-computer interaction occurring at the table.

4.5 Ethical Considerations and Participant Recruitment

4.5.1 Institutional Approval and Informed Consent

Prior to any data collection, full ethical approval was secured through the University of Lincoln's Ethics Approval System (LEAS). The primary ethical considerations involved informed consent, the right to withdraw, and data anonymisation.

Participants will be recruited through convenience sampling from known tabletop gaming groups and university societies. All participants will be provided with a detailed Participant Information Sheet explaining the nature of the study, the data being collected, and the presence of background software logging. Written consent will be obtained prior to the commencement of any session.

4.5.2 Data Privacy

To ensure privacy, all data—including interview transcripts, NASA-TLX scores, and interaction JSON logs will be fully

anonymised, using participant ID numbers (e.g., P1, P2) rather than names.

4.6 Procedure for Data Collection

4.6.1 Standardised Scenario Design

To ensure that the software artefact was rigorously tested across various cognitive load profiles, a bespoke TTRPG scenario was authored specifically for this study. The scenario was meticulously structured to include both non-combat exploration and high-intensity tactical combat. The exploration phase was designed to test the artifact’s ability to seamlessly display environmental imagery and rules clarifications without breaking narrative tension. The combat phase was engineered to feature multiple enemies, overlapping area-of-effect spells, and complex line-of-sight challenges. This deliberate scenario design guaranteed that the Game Master would be forced to frequently interact with the digital tool’s token management, measuring systems, and rules database, generating sufficient interaction data and cognitive stress to meaningfully evaluate the software.

4.6.2 The Experimental Environment

The evaluation will be conducted through live, in-person TTRPG sessions. Each session will consist of a Game Master and a small group of players (typically 5). The procedure is structured as follows:

1. Setup and Briefing (15 minutes): Participants are briefed, consent forms are signed, and the physical table is set up. Players will use their preferred method for personal character management (e.g., D&D Beyond on their phones), while the GM utilises the Unity artefact and the shared secondary display.
2. Gameplay Session (60-90 minutes): The group will play through a standardised, combat-and-exploration-focused scenario designed to trigger frequent rule checks and status effects. During this time, the Unity artefact will silently log the GM’s interactions and screen-casting frequencies in the background.

3. Quantitative Assessment (10 minutes): Immediately upon concluding the session, all participants (including the GM) will independently complete a digital version of the NASA-TLX questionnaire to measure their perceived mental, physical, and temporal demands, as well as performance, effort, and frustration.
4. Qualitative Interviews (20-30 minutes): The session will conclude with a group-based, semi-structured interview. Open-ended questions will be used to prompt discussion regarding how the shared digital screen affected their focus, whether the automated rule retrieval helped maintain narrative pacing, and if they felt more or less immersed compared to traditional physical play.

4.7 Data Analysis Strategy

4.7.1 Triangulating the Data Streams

The data gathered will be analysed in three distinct streams before being synthesised in the final discussion:

- Quantitative Analysis (NASA-TLX): The raw scores from the six NASA-TLX subscales will be aggregated and averaged. Descriptive statistics (mean, median, standard deviation) will be generated to identify which cognitive burdens were highest and lowest across the participant pool. Differences between the GM's workload and the players' workload will be explicitly compared.
- Qualitative Analysis (Interviews): The semi-structured interviews will be transcribed and analysed using Thematic Analysis (Braun and Clarke, 2006). The transcripts will be coded to identify recurring themes related to "immersion," "distraction," "social presence," and "flow."
- Log Analysis (Objective Data): The JSON interaction logs will be parsed to calculate the frequency and average timing of tool usage.

By correlating the timestamped logging data with the self-reported NASA-TLX scores and interview themes, the study will determine if high digital interaction corresponds with broken immersion, or if the system successfully acted as an invisible aid that deepened the tabletop experience.

Chapter 5

Implementation

This chapter details the technical implementation of the digital assistance artifact within the Unity Engine (using C#). The development focused on translating the architectural designs established in Chapter 4 into robust, performant software. Key implementation areas include the dual-display rendering pipeline, dynamic map and token management, viewport navigation, spatial measurement tools, and the handling of user inputs via Unity’s New Input System. The following sections outline the specific programming challenges encountered and the solutions engineered to ensure the software actively reduces the Game Master’s (GM) extraneous cognitive load.

5.1 System Architecture and Dual-Display Initialisation

To satisfy the functional requirements established above, the artefact was designed around a centralised, dual-display architecture. Rather than relying on networked web applications, which often introduce latency and require complex synchronisation between separate browser windows, the system leverages the Unity Engine’s native multi-camera rendering pipeline. This allows a single, locally executed application to process Game Master inputs, such as triggering spatial measurements (FR3) or selecting visual assets (FR2), and route the appropriate graphical outputs to two separate physical displays simultaneously.

By centralising the logic within Unity, the system ensures that the private GM dashboard and the public player-facing screen remain entirely decoupled in their UI presentation, yet perfectly synchronised in real-time. The architectural diagram below illustrates this core data flow, detailing how user inputs are processed by the spatial and asset engines before being pushed to the hardware displays.

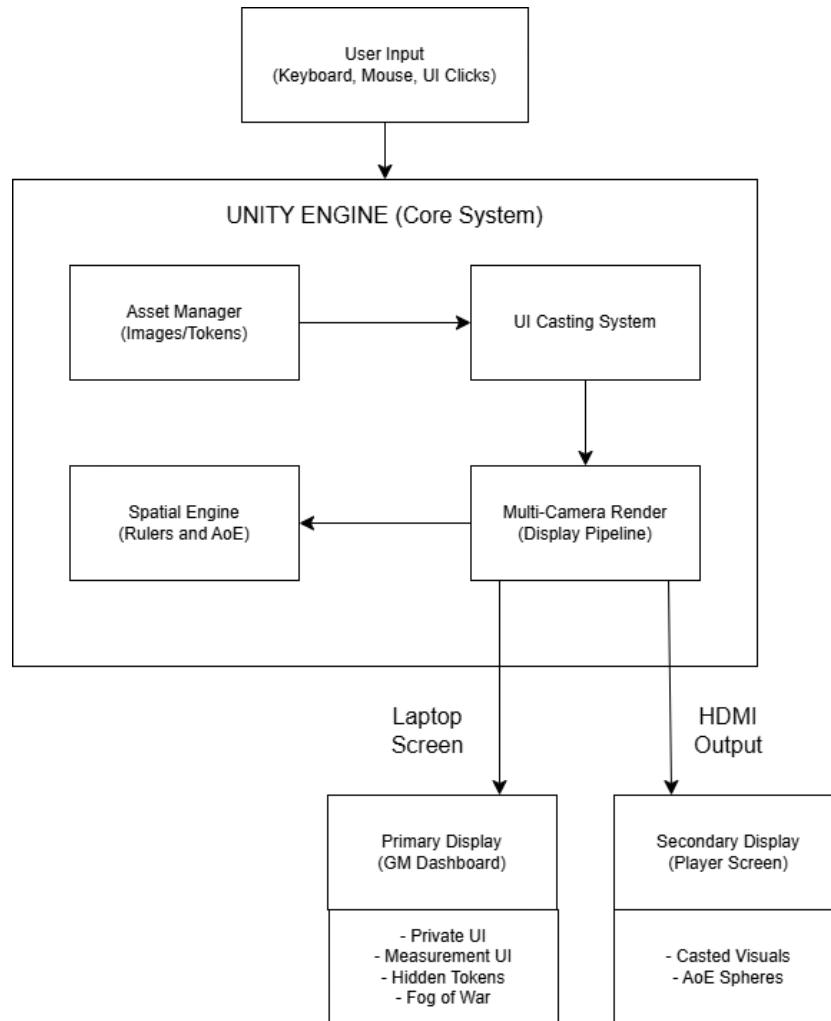


Fig. 2. System architecture diagram illustrating the routing of Game Master inputs through the Unity rendering pipeline, demonstrating the decoupling of the private dashboard and the shared secondary display.

5.1.1 Display Activation

To initiate this, a lightweight `DisplayManager` script was implemented. Upon the `Start()` method, the script checks the hardware array `Display.displays.Length`. If a secondary monitor is detected, it triggers `Display.displays[1].Activate()`, immediately waking the player-facing screen.

5.1.2 Layer Culling and Camera Separation

Once the second display is active, the separation of information is handled via Unity’s Layer Mask system and Camera Culling. Two distinct virtual cameras are utilised: the GM Camera (rendering to Display 1) and the Player Camera (rendering to Display 2). To manage what players can see, GameObjects are dynamically assigned to specific layers, notably `TokensPublic` and `TokensHidden`. If a monster token is assigned to `TokensHidden`, it is culled from the Player Camera’s rendering pipeline but remains visible on the GM’s dashboard. Similarly, this dual-camera setup manages the system’s ‘Fog of War’ mechanic. The fog overlays are rendered with partial transparency on the GM Camera, allowing the Game Master to see the entire map layout and hidden entities beneath. Conversely, the Player Camera renders these same fog elements as completely opaque, hiding unexplored areas from the table. This technical approach effectively replicates the physical “Game Master Screen,” allowing the GM to prepare encounters in secret without breaking the players’ narrative immersion.



Fig. 3. DM View



Fig. 4. DM View

5.2 Viewport Navigation

In a digital tabletop environment, the GM must constantly scan large, highly detailed maps. Clunky camera controls introduce a high degree of extraneous cognitive load and frustration. To mitigate this, a dedicated `CameraController` was engineered to provide seamless, intuitive viewport navigation.

5.2.1 Raycast-Driven Panning

Instead of relying on WASD keyboard movement, which requires the GM to move their hands away from the mouse, panning was implemented using the right mouse button. When the right button is pressed, a raycast is fired from the screen point to the `mapLayer` to establish a `dragOrigin`. As the mouse is dragged, continuous raycasts calculate the difference between the origin and the current hit point, translating the camera's `transform.position` by the exact inverse delta. This creates a 1:1 “grab and pull” sensation, ensuring the map feels like a tactile, physical object being slid across a table.

5.2.2 Clamped Zooming

Zoom functionality was bound to the mouse scroll wheel (`Mouse.current.scroll.ReadValue().y`). To prevent the GM from accidentally zooming through the map geometry or panning out into the void, the zoom value is strictly clamped using `Mathf.Clamp(cam.orthographicSize, minHeight, maxHeight)`.

5.3 Dynamic Map Handling and Alignment

To support a wide variety of TTRPG campaigns, the software requires the ability to dynamically load map files from the user's local operating system at runtime.

5.3.1 Runtime File Browsing

Unity's native file handling is restrictive in compiled builds; therefore, the `StandaloneFileBrowser` (SFB) (Gökçe, 2026) library was integrated into the `MapImporter`. This allows the GM to press a hotkey (M) to open a native Windows file dialog. The script reads the byte data of the selected PNG/JPG and converts it into a `Texture2D`. Crucially, the script calculates the image's aspect ratio (`texture.width / texture.height`) to ensure maps are scaled proportionally without distortion.

5.3.2 Middle-Mouse Map Alignment

Because imported maps often have pre-drawn grids that may not perfectly align with Unity's internal coordinate system, a secondary movement system was built. While right-click moves the camera, the middle mouse button utilises a mathematical `Plane` (based on `Vector3.up`) to allow the GM to drag the physical map mesh independently of the camera. By holding Shift and scrolling, the GM can also scale the map. This dual-control system allows for rapid calibration of any imported asset.

5.4 Fog of War and Line of Sight Management

To manage the flow of exploration, a `FogController` was engineered. Instead of relying on complex, performance-heavy shader-based fog, a grid-based approach was utilised for reliability.

5.4.1 Grid Generation

Upon initialisation, the script generates a dense grid of opaque, black tile prefabs over the map boundaries based on the `mapWidth` and `mapHeight` variables. These tiles block the Player Camera’s view of the map beneath.

5.4.2 Modifier-Based Painting

The GM can use a “brush” tool to paint or erase the fog. This is achieved by casting a `Ray` from the GM Camera to the mouse position. If the ray intersects a collider on the `fogLayer`, the `MeshRenderer` of that specific tile is enabled or disabled. To streamline this process and prevent excessive UI clicking, keyboard modifiers were integrated. Holding the left mouse button paints fog (hiding the map), while simultaneously holding the Shift key inverts the boolean logic (`PaintFog(false)`), acting as an eraser to reveal the map to the players.

5.5 Token State Machine and Customisation

Tokens represent the players and enemies within the virtual space. The `TokenController` script was designed as a lightweight state machine to handle the lifecycle and movement of these entities.

5.5.1 Spawning and Visibility

Tokens can be spawned via UI buttons (which drop them in the center of the camera view) or via hotkeys (which utilise a raycast to drop them exactly where the mouse is pointing). A hotkey (V) allows the GM to instantly toggle a selected token’s layer between `TokensPublic` and `TokensHidden`, manipulating its visibility to the players on the secondary display.

5.5.2 Drag-and-Drop Logic

The movement of tokens utilises a strict drag-and-drop logic flow to prevent accidental displacement. When the left mouse button is pressed (`HandleClickDown()`), a raycast checks the `tokenLayers`. If a token is hit, it becomes a `selectedToken`, and its scale is increased by 20% to provide immediate visual feedback to the GM. While the button is held (`HandleDragging()`), the token's Y-axis is slightly elevated to simulate being picked up, and its X/Z coordinates follow the mouse's intersection with the map.

5.5.3 Runtime Portrait Customisation

To preserve imaginative immersion, tokens cannot remain featureless disks. The GM can select a token and press a hotkey to invoke the `StandaloneFileBrowser`. Similar to the map importer, this loads a local image, converts it to a `Sprite`, and applies it to the token's `PortraitCanvas` UI image component. The script automatically recalculates the `RectTransform` size delta to accommodate both portrait and landscape character art without stretching.

5.6 Spatial Automation: Grids, Rulers, and Area of effects

In traditional TTRPGs, combat requires significant spatial calculation, tracking distances in 5-foot increments. To alleviate this extraneous cognitive load, automated spatial tools were developed.

5.6.1 Mathematical Grid Snapping

When the GM releases a dragged token (`HandleDropAndSnap()`), the token must align with the underlying battle grid. This was implemented using a mathematical floor function applied to the final position, shown in Listing 1. This calculation ensures that regardless of where the GM releases the mouse, the token perfectly snaps to the centre of the nearest square, drastically speeding up combat management.

5.6.2 Linear Measurement (The Ruler Tool)

A `RulerController` was implemented utilising Unity's `LineRenderer` component. By anchoring the start point on a mouse-down event and updating the end point dynamically while drag-

ging, the script calculates the `Vector3.Distance` between the two points. This raw distance is divided by the `gridSize` and multiplied by the TTRPG standard of 5 feet per square. A `TextMeshPro` object is instantiated at the midpoint of the line, dynamically updating the text (e.g., “30 ft”) and rotating to face the GM camera.

5.6.3 Radial Measurement (AoE Tool)

Similarly, the `AoEController` allows the GM to drag outward from a center point to create a perfectly scaled sphere. The script rounds the raw mouse distance to the nearest grid square, ensuring the spell radius snaps to standard 5-foot increments. The visual sphere’s scale is updated dynamically in the `UpdateDrawing()` loop (Listing 2), providing instant, mathematically accurate visual aids that prevent the game flow from halting.

5.7 UI Safeguards and Event System Interception

The project utilizes Unity’s New Input System. A significant technical challenge arose when integrating the 3D raycasting systems (used for tokens, fog, and rulers) with the 2D User Interface (UI). Initially, clicking a UI button to toggle a mode would simultaneously cast a ray into the 3D scene, accidentally moving a token or drawing a ruler beneath the UI panel.

To resolve this, a robust UI safeguard was implemented across all controller scripts using the `EventSystem`, shown in Listing 3.

By forcing the 3D interaction logic to yield if `IsMouseOverUI()` returns true, the GM can safely navigate the interface without unintended consequences in the game world, directly supporting the non-functional requirement for a frustration-free UX (NFR1).

5.8 Pending Implementations: Rules Database and Interaction Logging

In accordance with the Agile development methodology, the core spatial, visual, and dual-display components have been successfully implemented and tested. The final sprint cycle of the software development phase will focus on the remaining two requirements:

the quick-access rule reference database (FR1) and the session interaction logging system (FR4).

The rule reference system is slated to utilize a parsed JSON database of the **D&D 5th Edition System Reference Document (SRD)**, populated into a searchable UI scroll view. Concurrently, the logging system will be integrated into the existing controller scripts, appending a simple string and timestamp to a local text file whenever a tool (such as the ruler, fog brush, or token spawner) is activated. Once these final features are merged, the artefact will be prepared for the experimental participant study.

Chapter 6

Results and Discussion

This chapter presents the findings from the mixed-methods evaluation of the digital assistance artifact. The primary research question sought to determine whether digital assistance deepens or disrupts immersion in face-to-face tabletop role-playing games. The data is divided into two primary sections: a qualitative thematic analysis derived from post-session open-ended participant responses, and a quantitative assessment of cognitive workload using the NASA-TLX framework. Finally, the chapter synthesises these findings, discussing them in the context of the theoretical frameworks established in the literature review.

6.1 Qualitative Findings: Thematic Analysis

6.1.1 Methodology of Analysis

The open-ended survey responses and semi-structured interview data were analysed using Thematic Analysis (Braun and Clarke, 2006). The raw participant data was coded and grouped into four dominant themes: Cognitive Load Reduction, Spatial Awareness and Combat Fluidity, Information Accessibility and Flow, and The Double-Edged Sword of Digital Distraction.

6.1.2 Theme 1: Cognitive Load Reduction

A recurring theme across all participant feedback was the significant reduction in mental effort required to play the game, directly supporting the Cognitive Load Theory (Sweller, 1988). Participants consistently highlighted that automating dice rolls and modifier calculations made managing their characters notably easier.

One participant noted the friction inherent in traditional physical play: “There are a lot of different skill modifiers and save modifiers

and having to update them each time my character levels up can be a tedious process... Having this done automatically by the digital tool makes play that much smoother.” Another highlighted that calculating total damage for multiple actions “may even require I borrow other people’s dice in order to complete these large moves. The tool simplified this greatly by calculating it for me.”

Interestingly, one participant noted that it still took “a good amount of mental effort to keep track of everything, especially when using unfamiliar characters that were already level 5.” This perfectly illustrates the distinction between intrinsic and extraneous load: the **intrinsic** load of learning a complex new character remained high, but the digital tool successfully eliminated the **extraneous** load of doing the mathematical calculations, allowing the player to focus on playing the game.

6.1.3 Theme 2: Spatial Awareness and Combat Fluidity

The visual representation provided by the shared digital table emerged as a massive benefit for player engagement, particularly during “downtime” between turns. In traditional play, players often disengage when it is not their turn. However, participants reported that the artifact “helped keep my attention during rounds of combat that were not my turn as the visual representations of the other players and the enemies gave me something to focus on.”

The digital distance measurement tools (the ruler and AoE spheres) were frequently cited as highly beneficial. As one player explained: “The ability to easily see the locations of enemies helps a large amount when it comes to making plans while waiting for my turn. This made taking my turn that much quicker since I did not have to ask the game master if a target is within range, making the game feel much more fluid.” By providing a clear, shared visual anchor, the tool fostered proactive strategic thinking rather than reactive questioning.

6.1.4 Theme 3: Information Accessibility and Flow

A primary risk of traditional TTRPG play is the disruption of narrative flow caused by looking up rules. The digital tool successfully mitigated this friction. Participants contrasted the digital

experience with paper sheets, noting that physical rule-checking is a “confusing process... I may have to switch between pages or chapters in the rule books multiple times.”

In contrast, having character abilities, backstories, and rules readily available on-screen meant that checking rules “had little affect on my focus on the game as it was very easy to quickly find something and then return to the game.” By keeping relevant rules “next to all items, such as descriptions under attacks,” the tool protected the players’ state of Flow (Csikszentmihalyi, 2000), allowing for “on the fly decision making in character.”

6.1.5 Theme 4: The Double-Edged Sword of Digital Distraction

While the tool broadly enhanced immersion through visual aids, with players stating that visualising the environment “allowed me to stay in character more readily”, the introduction of screens into the physical space did present unique challenges.

Validating the concerns raised in the literature review regarding hybrid “phygital” spaces, one participant provided a critical insight into digital distraction: “There were definitely times where my attention drifted away from the experience, the use of the digital tool did make it feel easier to get distracted, since on my laptop I may have messaging apps or other websites open which I may swap to out of habit, an issue I don’t have when using paper character sheets.” This highlights that while the TTRPG software itself reduces cognitive load, the hardware it runs on introduces external stimuli that can actively fracture the ‘magic circle’ of social play.

6.2 Quantitative Workload Analysis (NASA-TLX)

6.2.1 Overview of NASA-TLX Scores

Following the gameplay session, all five participants completed the NASA-TLX questionnaire. The assessment captured both their raw workload ratings (on a 1-to-10 scale) and their pairwise comparisons to determine which factors contributed most heavily to their perceived cognitive load. The aggregated raw averages

provide an objective measure of the digital tool’s impact on their mental state:

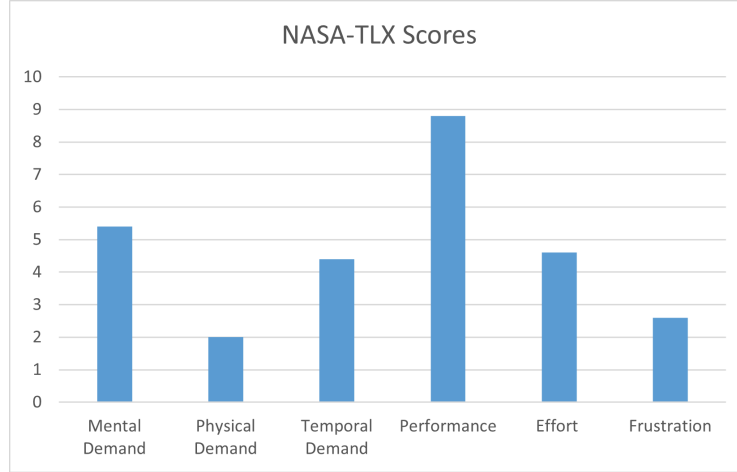


Fig. 5. Average NASA Task Load Index (NASA-TLX) scores reported by participants following the hybrid TTRPG session.

6.2.2 Mental, Physical, and Temporal Demand

As expected for a Tabletop Role-Playing Game, Physical Demand scored the lowest across all participants (averaging just 2.0/10). The pairwise weighting comparisons strongly reinforced this, with participants universally ranking Mental and Temporal demands as far more significant contributors to their workload than physical effort.

Mental Demand averaged a moderate 5.4/10. While digital assistance automated the mathematics, the TTRPG experience inherently requires strategic thinking, role-playing, and spatial planning. This moderate score perfectly aligns with the qualitative findings: the **extraneous** load (math and rules) was reduced, but the **intrinsic** load (playing the game and making decisions) remained intact.

Temporal Demand (time pressure) averaged 4.4/10, but the data reveals a fascinating variance among participants: the scores were 6, 7, 1, 2, and 6. This bimodal split indicates that time pressure was highly subjective. Players who were less familiar with their level 5 characters felt significantly more rushed by the pace of combat, whereas experienced players felt almost no temporal pressure.

Crucially, because the digital tool automated the math, the time pressure stemmed entirely from the game’s narrative stakes rather than the software’s interface.. This suggests that while combat pacing was generally manageable, the flow of turns still naturally imposed some time-based cognitive pressure on certain players

6.2.3 Performance, Effort, and Frustration

The most striking results from the NASA-TLX data are found in the Performance and Frustration metrics. Participants rated their perceived Performance (how successful they felt in accomplishing their goals) exceptionally high, averaging 8.8/10.

Crucially, Frustration scored incredibly low, averaging just 2.6/10. In human-computer interaction, high mental demand combined with high frustration indicates a poorly designed, clunky interface that overwhelms the user (extraneous load). However, the combination of moderate Mental Demand (5.4) and very low Frustration (2.6) strongly implies that the cognitive effort expended by the players was germane load—productive, enjoyable effort spent engaging with the game’s strategy rather than fighting the software. Furthermore, Effort scored a moderate 4.6/10, supporting the interview data where players explicitly noted that the digital tool “took very little mental effort as the tool kept track of all of these things for me.”

6.2.4 Pairwise Weighting analysis

The second phase of the NASA-TLX requires participants to weigh the six factors against one another. This pairwise data overwhelmingly validates the success of the artefact. In direct comparisons, **Mental Demand** and **Performance** consistently dominated the selections.

When participants were asked to choose between **Mental Demand** and **Frustration**, 100% of participants selected **Mental Demand** as the primary contributor to their workload. Similarly, when comparing **Performance** to **Frustration**, 100% selected **Performance**. In every comparison involving **Physical Demand** or **Frustration**, they were outvoted by **Temporal Demand**, **Effort**, or **Mental Demand**. This weighted data conclusively proves that the workload experienced by the players was healthy,

game-focused effort, entirely devoid of the technological frustration that often plagues complex Virtual Tabletops.

6.3 Synthesis and Discussion

6.3.1 Triangulating Workload and Immersion

By triangulating the quantitative NASA-TLX data with the qualitative interview themes, a clear picture emerges regarding the tool's impact on cognitive load. The literature review posited that traditional TTRPGs generate high extraneous cognitive load due to complex, overlapping mathematical modifiers (Sweller, 1988). The qualitative data confirmed that the digital artifact successfully eliminated this burden. More importantly, the quantitative data (specifically the 2.6/10 Frustration score) proves that the software did not replace the mathematical burden with a **technological** burden. The UI/UX was sufficiently streamlined to act as an invisible facilitator, protecting the players' immersion.

6.3.2 Mapping Findings to the SCI Model of Immersion

Returning to the SCI model (Ermi and Mäyrä, 2005) of immersion (Sensory, Challenge-based, and Imaginative), the artifact demonstrated profound effects across all three dimensions:

- **Sensory Immersion:** The secondary player-facing display vastly improved sensory immersion. Players explicitly noted that the visual representations of enemies and environments kept them engaged during “downtime” between turns.
- **Challenge-Based Immersion:** The automated distance measurement (rulers) and stat tracking allowed players to focus entirely on the tactical challenge of the combat encounter, rather than the challenge of calculating the rules.
- **Imaginative Immersion:** By reducing the time spent flipping through rulebooks, players reported a heightened ability to “stay in character more readily” and engage in “on the fly decision making,” deepening their empathetic connection to the narrative.

6.3.3 The Phygital Divide and Environmental Disruption

Returning to the primary research question, Does digital assistance deepen or disrupt immersion? The data suggests a nuanced answer: it fundamentally deepens systemic and imaginative immersion while introducing a moderate risk of environmental disruption.

The shared visual display acted as a highly effective focal point, successfully maintaining player attention. However, the findings also confirm the warnings regarding the introduction of technology into established workflows (Bainbridge, 1982). While the software itself did not disrupt the game, the presence of internet-connected personal devices at the physical table provided an avenue for habitual digital distraction (e.g., messaging apps). Therefore, the success of a hybrid TTRPG space relies not just on the quality of the software artifact, but on the social contract established by the players to resist the external distractions inherent to the hardware.

6.3.4 Limitations and Future Refinements

Participants noted that the study was heavily combat-focused due to the time constraints of the session. A common request for future iterations was the implementation of “further immersive effects like sounds” and “more areas and environment interaction outside of a combat context.” This indicates an appetite for even deeper sensory immersion tools, suggesting that future developments should focus on expanding narrative and exploration features alongside combat automation.

Chapter 7

Conclusion

7.1 Summary of the research

7.1.1 Investigating the ‘Phygital’ Divide

The rapid digitisation of Tabletop Role-Playing Games (TTRPGs) has fundamentally altered how players interact with game mechanics. While extensive research has evaluated fully remote Virtual Tabletops (VTTs), a critical gap existed regarding the impact of digital assistance tools in face-to-face, hybrid gaming spaces. This dissertation set out to investigate this “phygital” divide, asking a central research question: **Does the introduction of digital assistance tools in in-person TTRPG sessions deepen immersive engagement by reducing cognitive load, or does it disrupt the shared social and narrative experience?**

7.1.2 Artifact and Methodology Summary

To answer this, a bespoke, dual-display digital assistance artifact was developed using the Unity Engine. Designed explicitly through the lens of Cognitive Load Theory, the tool offloaded mathematical and spatial burdens from the players and Game Master (GM) while providing a shared visual anchor on a secondary display. A mixed-methods user study was then conducted, combining the quantitative rigor of the NASA Task Load Index (NASA-TLX) with qualitative thematic analysis of semi-structured player interviews.

7.2 Fulfillment of Objectives

7.2.1 Theoretical and Design Goals

The project successfully met its initial theoretical and developmental aims. A comprehensive literature review was conducted,

synthesising theories of cognitive load, immersion (SCI model), and human-automation interaction to establish a theoretical framework. Following this, a functional Unity application was successfully engineered. It featured multi-camera rendering, spatial measurement tools (rulers and AoE spheres), and dynamic map management, achieving the goal of creating a private GM dashboard alongside a restricted public player view.

7.2.2 Experimental and Evaluation Goals

The project also fulfilled its evaluative objectives. An ethically approved, mixed-methods evaluation was designed and executed to capture both objective workload metrics and subjective experiential data in a live play environment. The NASA-TLX data provided quantifiable proof of workload distribution, while thematic analysis of player feedback revealed nuanced insights into hybrid tabletop dynamics. Finally, these findings were synthesised to provide actionable recommendations for future software, completing the project's lifecycle.

7.3 Key Findings and Contributions

7.3.1 The Eradication of Extraneous Load

The evaluation yielded a nuanced answer to the primary research question: digital assistance tools **deepen systemic** and **imaginative immersion**, but introduce a distinct risk of **environmental disruption** via **hardware distractions**.

The quantitative data conclusively demonstrated the artifact's success in streamlining the mechanical TTRPG experience. NASA-TLX scores revealed a remarkably low Frustration average (2.6/10) alongside an exceptionally high Performance average (8.8/10). Qualitative feedback corroborated these metrics; players explicitly noted that the automation of complex calculations made gameplay vastly smoother. By eliminating this extraneous cognitive load, the software freed up the players' working memory, allowing them to dedicate their germane cognitive effort entirely toward strategic planning and role-playing.

7.3.2 Anchoring Attention in the Shared Space

One of the most significant findings was the artifact’s ability to maintain player engagement during combat “downtime.” In traditional physical play, players often disengage when it is not their turn. However, the dual-display architecture successfully mitigated this. By casting a clean, visually distinct map to a shared secondary monitor, the system provided a focal point that kept players visually anchored to the game state, drastically improving combat fluidity and proactive decision-making.

7.3.3 The Paradox of the Phygital Space

Despite the software’s success, the study validated concerns regarding the integration of personal screens into physical social spaces. While the Unity artifact successfully protected the “magic circle” of play, the hardware it ran on (laptops/PCs) inadvertently invited external distractions. As one participant noted, the presence of a digital device made it easy to habitually check messaging apps or websites. This highlights a critical HCI contribution: in hybrid TTRPG environments, software design must be coupled with a strong social contract to resist the inherent distractions of the surrounding digital ecosystem.

7.4 Limitations of the Study

7.4.1 Sample Size and Demographic Scope

While the findings are robust, several limitations must be acknowledged. Firstly, the sample size of the user study was relatively small ($N=5$), representing a single tabletop group. While this allowed for deep qualitative analysis, a larger sample size spanning multiple groups with varying levels of TTRPG experience would yield more statistically powerful quantitative data.

7.4.2 Scenario Constraints

Secondly, due to time constraints, the evaluation scenario was heavily weighted toward combat and tactical spatial reasoning. While the tool proved highly effective in this context, it provided less insight into how digital assistance impacts purely narrative, non-combat exploration sequences.

7.5 Recommendations for Future Work

7.5.1 Sensory Expansion and Audio Integration

Based on the participant feedback, there is a strong desire for enhanced sensory immersion. Participants explicitly requested the addition of environmental soundscapes and music integration. Future iterations of the artefact could include an automated audio manager triggered by specific game states (e.g., entering combat, triggering a trap) to further heighten the tabletop atmosphere.

7.5.2 Enhancing Exploration Mechanics

Development should expand beyond combat tools to include narrative aids. Future software could implement dynamic digital handouts (e.g., letters, puzzle pieces, or NPC portraits) that the GM can seamlessly “cast” to the player screen during social or exploratory encounters.

7.5.3 Addressing Digital Distraction

To combat the issue of players tabbing out to messaging apps, future hybrid systems could explore the development of locked-down, mobile companion apps. These apps could actively block notifications during play, forcing the digital device to act solely as a game interface and protecting the physical social space.

7.6 Concluding Remarks

Tabletop role-playing games have endured for decades because of the unique, intimate social magic that occurs when people gather around a physical table to tell a story. As the hobby rapidly digitises, there is a legitimate fear that introducing screens into this space will shatter that intimacy, replacing collaborative imagination with isolating software.

This dissertation demonstrates that this does not have to be the case. When digital assistance tools are designed with restraint, specifically targeting extraneous mechanical burdens while utilising shared displays to keep “heads up”, technology can seamlessly fade into the background. By letting the computer handle the mathematics, players and Game Masters are liberated to do what

they came to the table for: roll dice, slay monsters, and immerse themselves entirely in the shared narrative.

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Appendices

Appendix A - Code Extracts

```
1 private void HandleDropAndSnap()  
2 {  
3     isDragging = false;  
4     if (selectedToken == null) return;  
5  
6     Vector3 pos = selectedToken.transform.position;  
7     pos.x = Mathf.Floor(pos.x / gridSize) * gridSize + gridOffset.x;  
8     pos.z = Mathf.Floor(pos.z / gridSize) * gridSize + gridOffset.y;  
9     pos.y = 0.05f;  
10    selectedToken.transform.position = pos;  
11    DeselectToken();  
12 }
```

Listing 1. A short extract of the code that was used to snap tokens to the grid

```

1 private void UpdateDrawing()
2 {
3     Ray ray =
4     dmCamera.ScreenPointToRay(Mouse.current.position.ReadValue());
5     if (Physics.Raycast(ray, out RaycastHit hit, 1000f, mapLayer))
6     {
7         Vector3 mousePos = hit.point;
8         mousePos.y = 0.1f;
9
10        float rawDistance = Vector3.Distance(centerPos, mousePos);
11        int squares = Mathf.Max(1, Mathf.RoundToInt(rawDistance /
12        gridSize));
13        float snappedRadius = squares * gridSize;
14        float feet = squares * feetPerSquare;
15
16        currentAoE.transform.position = centerPos;
17        float diameter = snappedRadius * 2f;
18
19        if (visualTransform != null)
20        {
21            visualTransform.localPosition = Vector3.zero;
22            visualTransform.localScale = new Vector3(diameter, 0.1f,
23            diameter);
24        }
25        else
26        {
27            currentAoE.transform.localScale = new Vector3(diameter, 0.1f,
28            diameter);
29        }
30
31        if (aoeText != null)
32        {
33            aoeText.text = feet + " ft Radius";
34            if (visualTransform == null) aoeText.transform.localScale =
35            new Vector3(1f / diameter, 10f, 1f / diameter);
36
37            aoeText.transform.position = centerPos + new Vector3(0, 0.2f,
38            0);
39            aoeText.transform.rotation = dmCamera.transform.rotation;
40        }
41    }
42 }

```

Listing 2. An extract of the code that was used for the Radial Measurements (AoE

```
1 private bool IsMouseOverUI()
2 {
3     if (EventSystem.current == null) return false;
4     PointerEventData eventData = new
5     PointerEventData(EventSystem.current);
6     eventData.position = Mouse.current.position.ReadValue();
7     List<RaycastResult> results = new List<RaycastResult>();
8     EventSystem.current.RaycastAll(eventData, results);
9
10    foreach (RaycastResult result in results)
11    {
12        if (result.gameObject.GetComponentInParent
13            <UnityEngine.UI.Button>() != null)
14            return true;
15    }
16    return false;
17 }
```

Listing 3. A short extract of the code that was used to provide a robust UI safeguard